ASSIGNMENT-1 : INPUT & OUTPUT FUNCTIONS

1. Print "Hello World" on the screen.

In [21]:
```
print("Hello World")
```

```
Hello World
```

2. Input two numbers design a calculator using +, -, *, % and / operators.

In [22]:
```
a = int(input("Enter a :"))
b = int(input("Enter b :"))
add = a+b
sub = a-b
mul = a*b
div = a/b
mod = a%b

print("Addition : ", add)
print("Subtraction : ", sub)
print("Multiplication : ", mul)
print("Division : ", div)
print("Modulus : ", mod)
```

```
Enter a :5
Enter b :9
Addition :  14
Subtraction :  -4
Multiplication :  45
Division :  0.5555555555555556
Modulus :  5
```

3. Find the square root of a number.

In [25]:
```
a = int(input("Enter a :"))
sq = a**0.5
print(sq)
```

```
Enter a :8
2.8284271247461903
```

4. Input the temperature in Fahrenheit and change it to Celsius.

In [27]:
```
temp = int(input("Enter temperature in fahrenhieite : "))
celcius = (temp-32) * 5/9
print(celcius, "degree in celcius")
```

```
Enter temperature in fahrenhieite : 54
12.222222222222221 degree in celcius
```

5. Input the basic salary of a person and compute its TA (20% of Basic), DA (120% of Basic), HRA (30% of Basic), and Gross (Basic + TA + DA + HRA).

In [37]:
```
basic = int(input("Enter basic salary : "))
TA = 0.2 * basic
DA = 0.12 * basic
HRA = 0.3 * basic
gross = basic + TA + DA + HRA
print("TA=", TA)
print("DA=", DA)
print("HRA=", HRA)
print("Gross=", gross)
```

```
Enter basic salary : 10000
TA= 2000.0
DA= 1200.0
HRA= 3000.0
Gross= 16200.0
```

6. Evaluate the expression: (4x^{4}+3y^{3}-9z+6)

In [46]:
```
x = int(input("Enter x : "))
y = int(input("Enter y : "))
z = int(input("Enter z : "))
res = 4*x**4 + 3*y**3 - 9*z + 6
print("Result = ", res)
```

```
Enter x : 5
Enter y : 4
Enter z : 1
Result =  2689
```

7. Take an input in uppercase and change it to lowercase.

In [47]:
```
s = input("Enter sentence in uppercase : ")
sl = s.lower()
print(sl)
```

```
Enter sentence in uppercase : HELLO WORLD
hello world
```

8. Input the radius of a circle and print its area and perimeter.

In [49]:
```
import math
r = int(input("Enter radius : "))
a = math.pi * r**2
p = 2 * math.pi * r
print("Area : ", a)
print("Perimeter : ", p)
```

```
Enter radius : 5
Area :  78.53981633974483
Perimeter :  31.41592653589793
```

9. Input marks in 5 subjects of a student and print its average mark.

In [50]:
```
m1 = int(input("Enter mark in subject 1 : "))
m2 = int(input("Enter mark in subject 2 : "))
m3 = int(input("Enter mark in subject 3 : "))
m4 = int(input("Enter mark in subject 4 : "))
m5 = int(input("Enter mark in subject 5 : "))
```

```
avg = (m1+m2+m3+m4+m5)/5
print("Average marks : ", avg)
```

```
Enter mark in subject 1 : 52
Enter mark in subject 2 : 48
Enter mark in subject 3 : 98
Enter mark in subject 4 : 78
Enter mark in subject 5 : 71
Average marks :  69.4
```

10. Input a number and print its square, cube and fourth power.

In [53]:
```
num = int(input("Enter number : "))
s = pow(num, 2)
c = pow(num, 3)
f = pow(num, 4)
print("Square = ", s)
print("Cube = ", c)
print("Fourth power = ", f)
```

```
Enter number : 2
Square =  4
Cube =  8
Fourth power =  16
```

11. Input three sides of a triangle and print its area.

In [56]:
```
x = int(input("Enter x : "))
y = int(input("Enter y : "))
z = int(input("Enter z : "))
s = (x+y+z)/2
a = (s*(s-x)*(s-y)*(s-z))**0.5
print("Area = ", a)
```

```
Enter x : 5
Enter y : 2
Enter z : 4
Area =  3.799671038392666
```

12. Input the amount and compute the simple interest and compound interest.

In [59]:
```
p = int(input("Enter principal: "))
s = p * 0.6 * 8
c = p * pow(1 + 0.6, 8)
print("SI = ", s)
print("CI = ", c)
```

```
Enter principal: 5000
SI =  24000.0
CI =  214748.3648000001
```

SRABANI KHUNTIA, 24BCSB03

GROUP: E1, ROLL NO: 7